

SHANMUKH KOMMU

+91 9121622304 ◇ Hyderabad, Telangana

kommu.shanmukh@gmail.com ◇ [LinkedIn](#) ◇ [GitHub](#) ◇

Agentic AI Engineer specializing in LLM-based automation, workflow agents, and enterprise RAG systems, with strong collaboration and communication skills in cross-functional environments.

EDUCATION

B.Tech. Artificial Intelligence and Data Science — KL University Hyderabad

Graduated: 2026 | CGPA: 9.42

SKILLS

Programming	Python, SQL, Java, C
Frameworks	PyTorch, TensorFlow, Django, Flask, Scikit-learn
ML/AI	Large Language Models (LLMs), Langchain, RAG, Transformers (BERT, GPT, HuggingFace), Pandas, NumPy, MCP, Langraph, AutoGen, Prompt Engineering
Tools	Git, Docker, Jenkins, CI/CD, Linux, PostgreSQL, MySQL, Kafka, Kubernetes, Keycloak, Celery, Redis
Data Formats, APIs	JSON, REST APIs, Websockets

EXPERIENCE

AI Intern

Techsophy

09/2025 - Present

Hyderabad, Telangana

- Engineered a **multi-camera person re-identification system** that reduced false positive matches by 73% in crowded environments. Leveraged 512-dimensional facial embeddings (**InsightFace**) and motion-based tracking (**DeepSort**) to enable security teams to track individuals across **8+ CP Plus** camera feeds with 94% accuracy
- Automated manufacturing date and batch number extraction** from product labels in warehouses, and reducing QC processing time by 18 hours/week through an OCR to transformer-based LLM pipeline that achieved 96% accuracy on degraded/rotated text, eliminating manual data entry for 300+ daily inspections. (**Paddle OCR, GPT-4o-nano.**)
- Built a real-time health monitoring pipeline using **Apache Kafka** to ingest and process high-volume patient vitals, structuring metrics into JSON schemas to construct a **digital twin**. Developed a **RAG agent** that indexes doctor-recommended research journals as vectors in MongoDB and queries the digital twin to generate automated diagnostic suggestions. Implemented Redis for cache storage and containerized using Docker.

PROJECTS

- PulseAI** [\[GitHub\]](#)
 - Built PulseAI, a Kafka-powered personal activity intelligence system that tracks real-time window activity, streams events through Apache Kafka, persists data to SQLite, and generates AI-powered productivity insights using a local LLM (Mistral via Ollama). Developed a Tkinter desktop GUI for one-click start/stop and report generation. Implemented a Jenkins CI/CD pipeline with automated pytest testing on Docker.
- Hallucination Insurance** [\[GitHub\]](#)
 - Designed and deployed a **local-first Hallucination Insurance API** that detects LLM hallucinations using a hybrid semantic similarity and LLM verification approach. Automatically extracts and verifies individual claims from LLM responses against source documents using Sentence-Transformers and ChromaDB vector search, with Mistral (via Ollama) handling ambiguous cases. Built with FastAPI backend and Streamlit UI.

CERTIFICATIONS

AWS Certified AI Practitioner – Amazon Web Services

[Credential](#)

MongoDB Certified Associate Developer – MongoDB

[Credential](#)

GitHub Foundations Certification – GitHub

[Credential](#)